NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

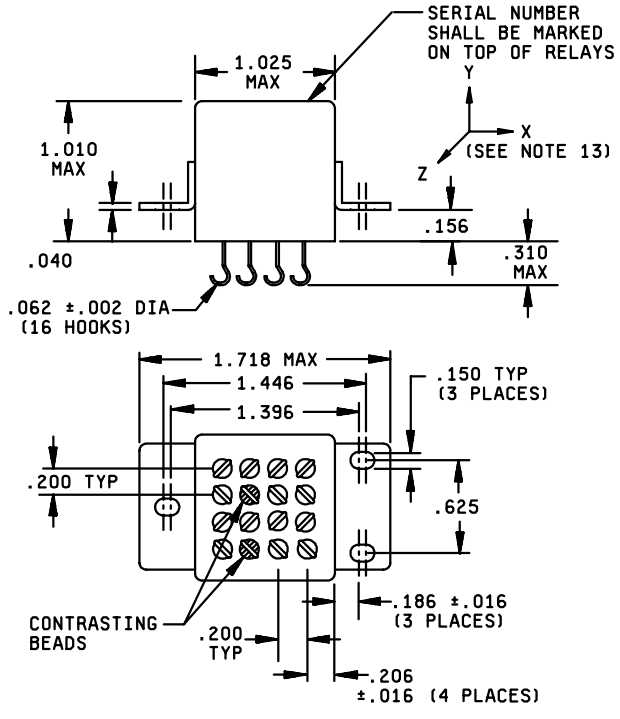
MIL-PRF-83536/20
27 March 1992

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, 4PDT, LOW LEVEL TO 10 AMPERES, MAGNETIC LATCH, HERMETICALLY SEALED, ALL WELDED, AC COILS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-PRF-83536.

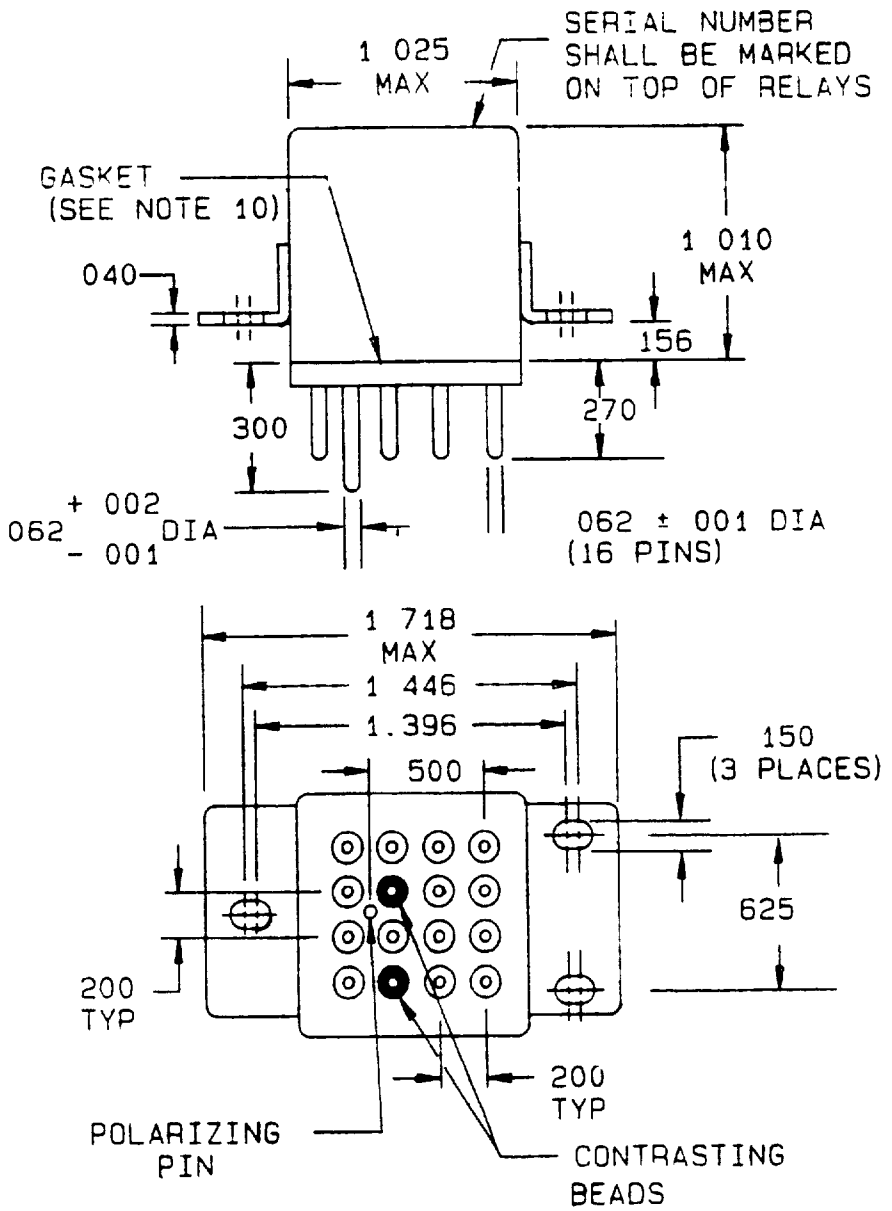



RAISED VERTICAL FLANGE MOUNT (RVFM) WITH SOLDER HOOKS

FIGURE 1. Dimensions and configurations.


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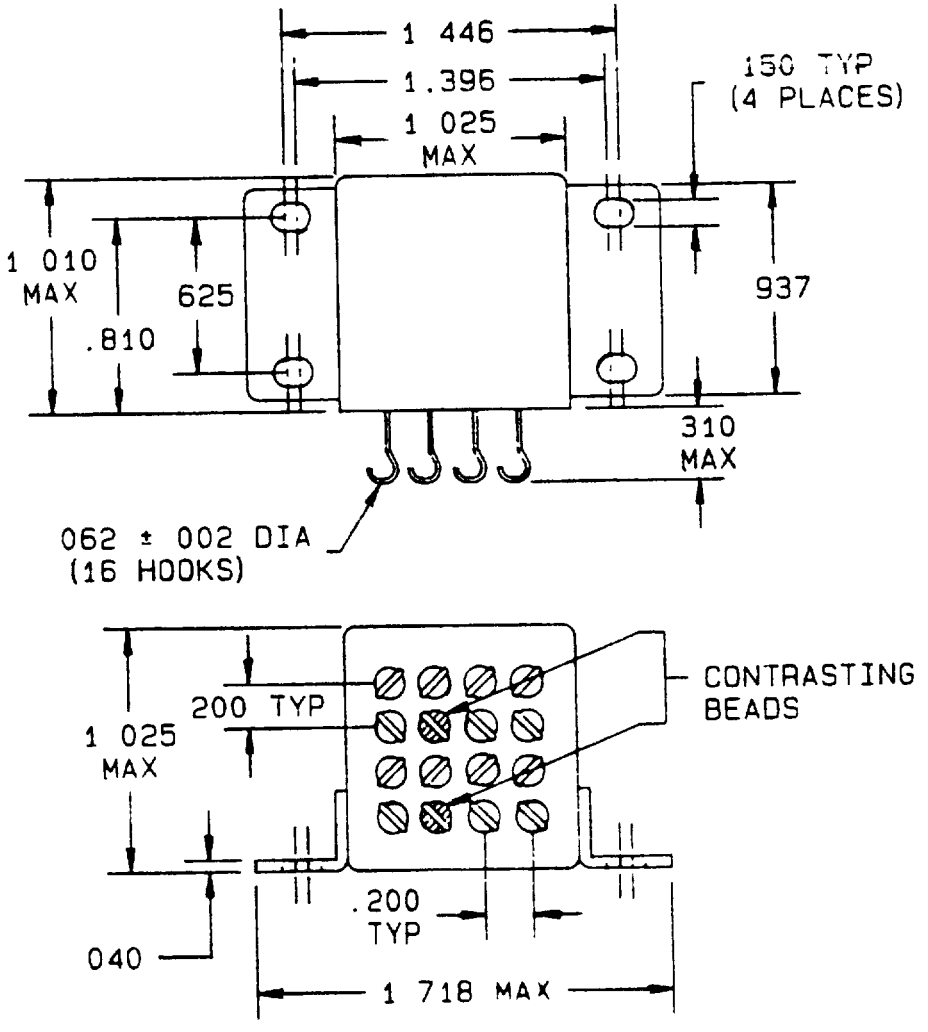



RAISED VETRICAL FLANGE MOUNT (RVFM) WITH SOCKET PINS

FIGURE 1 Dimensions and configurations - Continued



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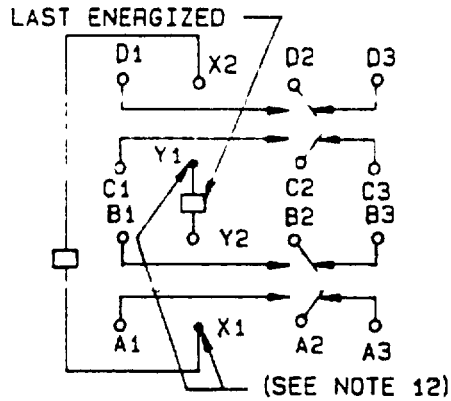



HORIZONTAL FLANGE MOUNT WITH SOLDER HOOKS

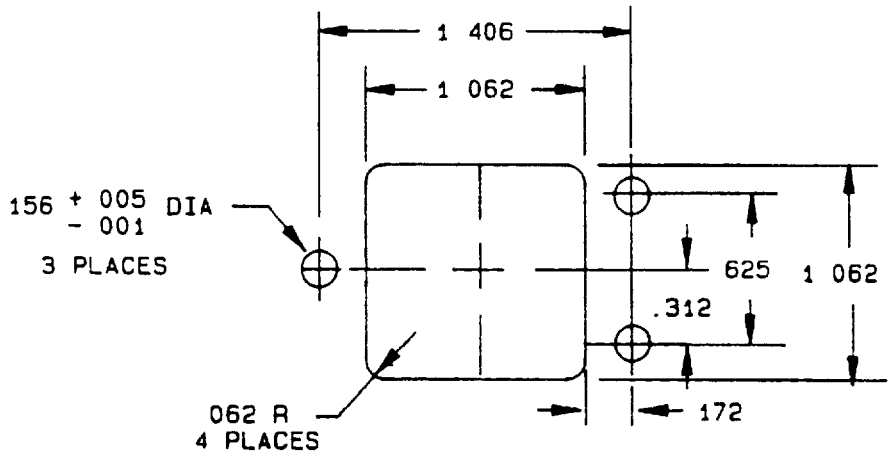
FIGURE 1 Dimensions and configurations - Continued



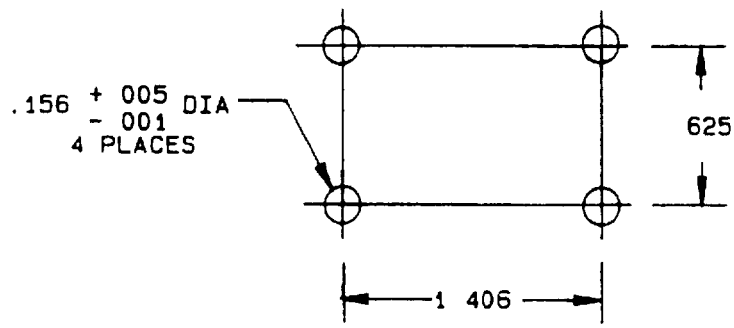
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CIRCUIT DIAGRAM




MOUNTING HOLE LAYOUT (RVFM)




MOUNTING HOLE LAYOUT (HFM)

FIGURE 1 Dimensions and configurations - Continued



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| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| 000 | 0 00 | 172 | 4 37 | 937 | 23 80 |
| 001 | 0 03 | .200 | 5 08 | 1 010 | 25 65 |
| 002 | 0 05 | 270 | 6.86 | 1 025 | 26 04 |
| 005 | 0 13 | 300 | 7 62 | 1.062 | 26 98 |
| 040 | 1 02 | 310 | 7 87 | 1 396 | 35 46 |
| 050 | 1 27 | .312 | 7 92 | 1 406 | 35 71 |
| 062 | 1 56 | .500 | 12 70 | 1.446 | 36 73 |
| 150 | 3 81 | 625 | 15 88 | 1 718 | 43 64 |
| 156 | 3 96 | 810 | 20 57 | | |

NOTES

1 Dimensions are in inches
2 Metric equivalents are given for general information only
3 Unless otherwise specified, tolerance is ±.010 (0.25 mm)
4 There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified
5 Permanent magnet drive consists of a permanent magnet with its flux path switched and combined with the electro-magnet flux
6 Applicable to horizontal flange mount only The circuit diagram, manufacturer's PIN, and the military PIN shall be marked on the near side The remaining portion of the nameplate data shall be marked on the far side
7 Applicable to horizontal flange mount only. Relays shall be marked with the manufacturer's name or CAGE code and date code Marking shall be with the bottom of the print adjacent to the near side.
8 Socket pin terminals, except the polarizing pin, shall be gold-plated in accordance with MIL-G-45204, type II, class I The nickel underplating shall be 50 to 150 microinches thick minimum
9 Silicone rubber gasket, AMS 3332, shore hardness 15 to 35, thickness 050 ± 005 (1 27 ±0 13 mm)
10 Track mount knob, post, and base shall be 303 Cres nickel plated in accordance with McAir P S 13112 or equivalent, 100 microinches thick minimum The latching pin shall be 303 Cres passivated or nickel plated in accordance with McAir P S 13112 or equivalent, 100 microinches thick minimum.
11 Direction of hook openings does not affect function
12 Indicated terminals shall be identified by contrasting beads
13 Plane of critical motion for vibration and shock is Y-axis

FIGURE 1 Dimensions and configurations - Continued



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REQUIREMENTS

Contact data

Load ratings See table I

Low level 10 to 50 µA at 10 to 50 mV dc or peak ac

Intermediate current Applicable

Mixed loads Applicable

TABLE I Rated contact load and cycles (amperes per pole)

| Type of load (high level) | Cycles x $10^3$ | 28 V dc | 115 V ac 1 phase 400 Hz | 115 V ac 1 phase 50/60 Hz 1/ | 115/200 V ac 3 phase 400 Hz | 115/200 V ac 3 phase 50/60 Hz 1/ |
|---|---|---|---|---|---|---|
| Resistive | 100 | 10 | 10 | 2 5 | 10 | 2 5 |
| Inductive | 20 | 8 | 8 | N/A | 8 | N/A |
| Inductive | 10 | N/A | N/A | 2 5 | N/A | 2 5 |
| Motor | 100 | 4 | 4 | 2 | 4 | 2 |
| Lamp | 100 | 2 | 2 | 1 | N/A | N/A |

1/ For 50/60 Hz rating, rupture and overload are not applicable and life for each load shall be 10,000 cycles

Life 100,000 cycles, unless otherwise specified (see table I)

Contact voltage drop or resistance

Initial contact voltage drop 0 100 volt maximum Rated resistive current at 6 V dc or peak ac Relays shall not make or break this load

Initial contact resistance 0 010 ohm maximum 50 mA max at 6 V dc or peak ac

High level life (contact voltage drop)

During life 10 percent open circuit voltage maximum

After life 0 125 volt maximum

Intermediate current (contact resistance)

During intermediate current 3 ohms maximum

After intermediate current 0 150 ohm maximum

Low level (contact resistance)

During low level 100 ohms maximum

After low level 0 150 ohm maximum at 100 mA and 28 V dc

Contact bounce 1 0 millisecond maximum

Contact stabilization time 2 5 milliseconds maximum

Overload current 40 amperes dc (60 amperes ac)



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Rupture current 50 amperes dc, 80 amperes ac

Time current relay characteristics See table II

TABLE II Time current relay characteristics 1/

| Successive application | Amperes | Time |
|---|---|---|
| 1 | 15 | 1 hour |
| 2 | 50 | 5 seconds |
| 3 | 100 | 1 2 seconds |
| 4 | 250 | 0 2 second |
| 5 | 350 | 0 1 second |

1/ All relays shall withstand overload and fault currents Relays must be able to sustain five applications (make and carry only) of power concurrently on adjacent poles at each of five different current levels in the sequence listed in table II. Separate relays shall be tested at 28 V dc and 115/200 V ac, 400 Hz 3-phase Cooling time between successive applications shall be 30 minutes Tests shall be performed on both normally open and normally closed contacts There shall be no failures or evidence of welding or sticking and relays shall pass contact voltage drop at the conclusion

Coil data 1/ See table III

Operate time 15 milliseconds maximum (each coil) with rated coil voltage over the temperature range

Duty rating. Continuous.

Coil transient suppression Applicable

Neutral screen Applicable

Electrical data

Insulation resistance

Initial 100 megohms minimum at 500 V dc

After life or environmental tests 50 megohms minimum at 500 V dc

1/ CAUTION Due to possible interaction of relay magnetic fields, the following spacing requirements, as a minimum, shall be considered in dense packaging situations.

a Row to row assisting fields 125 (3 18 mm)

b Row to row opposing fields 1875 (4 763 mm)

c Side to side alternating fields 0625 (1 588 mm)

d Side to side like fields 125 (3 18 mm)



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TABLE III Dash numbers and characteristics 1/

| Dash number 2/ | | Mount | Coil data | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | Coil voltage (V ac) 3/ | | Frequency (Hz) | Coil current (amperes) | Pickup voltage 4/ | | |
| Solder hook | Socket pin | | Rated | Max | | | Normal | High temperature test | Continuous current test |
| 001 | --- | RVFM | 115 | 122 | 400 | 04 | 90 | 95 4 | 103 5 |
| --- | 002 | RVFM | 115 | 122 | 400 | 04 | 90 | 95 4 | 103 5 |
| 003 | --- | HFM | 115 | 122 | 400 | 04 | 90 | 95 4 | 103 5 |

1/ Each relay possesses high level and low level capabilities However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits are not recommended for subsequent use in low level applications

2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number Failure rate level (percent per 10,000 cycles) L, 3 0, M, 1 0, P, 0 1, R, 0 01 Example 001L, 002R

3/ CAUTION The use of any coil voltages less than the rated coil voltage will compromise the operation of the relay

4/ Pickup voltage as shown is for test purposes only and is not to be used for design criteria

Dielectric withstanding voltage 2/

| Sea level | Coil to case | Coil to coil | All other points |
|---|---|---|---|
| Initial | 1,000 | 1,000 | 1,250 |
| After life | 1,000 | 1,000 | 1,000 |
| Altitude | | | |
| At 80,000 feet | 350 | 350 | 350 |
| At 300,000 feet | 500 | 500 | 500 |

Environmental characteristics

Temperature range -70°C to +125°C

Maximum altitude rating 300,000 feet

Shock (specified pulse) Applicable MIL-STD-202, method 213, test condition C, 200 g's for 6 ±1 ms, except horizontal flange mount peak g value shall be 100 g's for 6 ±1 ms Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts

2/ Dielectric may be improved by suitable insulation of terminals and wiring after installation



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Vibration (sinusoidal) Applicable, MIL-STD-202, method 204, 30 g's except frequency range shall be 10 to 3,000 Hz Horizontal flange mount shall be 20 g's, 57 to 3,000 Hz Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts

Vibration (random) Applicable to qualification and group C inspection Test in accordance with MIL-STD-202, method 214, test condition 1G (0.4 $g^2$/Hz, 50 to 2000 Hz), 15 minutes each plane, for horizontal flange mount test condition 1E (0.2 $g^2$/Hz Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts

Acceleration Applicable, except 15 g's

Physical data

Dimensions and configurations See figure 1

Weight 0.17 pound (77.2 g's)

Terminal strength

Solder hook terminals

Pull force 10 pounds, ±1 pound

Bend Applicable

Twist Not applicable

Socket pin terminals

Pull force 10 pounds, ±1 pound

Bend Applicable

Twist Not applicable

Terminal solderability Applicable to solder hook terminals only

Seal Hermetic

Marking Applicable

Part or Identifying Number (PIN) M83536/20 (dash number from table III and suffix letter designating failure rate level) PIN is a new term encompassing previous terms used in specifications such as part number, type designator, and identification number

Qualification inspection

Qualification inspection and sample size See table IV



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TABLE IV Qualification inspection and sample size

| Single submission | Group submission | |
|---|---|---|
| 18 units plus 1 spare unit for level L at C = 0 1/ | M83536/20-002 | 18 units plus 1 spare unit for level L at C = 0 1/ |
| 33 units plus 1 spare unit for level M at C = 0 1/ | " | 33 units plus 1 spare unit for level M at C = 0 1/ |
| Qualification inspection as applicable | " | Qualification inspection as applicable |
| | M83536/20-001 M83536/20-003 | 2 units, qualification inspection table, group II, shock, vibration, acceleration, terminal strength, and seal |

1/ The number of units required for qualification testing shall be increased as required in group V, table II of MIL-R-83536, if the contractor elects to test the number of units permitting one or more failures Prior to performance of qualification testing, the relay manufacturer shall preselect the sampling plan

Supersession data See table V

TABLE V Supersession data

| Superseded PIN M6106/51- | Replacement PIN M83536/20- |
|---|---|
| 007 | 001 |
| 008 | 002 |
| 009 | 003 |

CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Air Force - 99
DLA ES

Preparing activity
Air Force - 85

Agent
DLA - ES

(Project 5945-0857)



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